

Program : Diploma in Civil and Environmental Engineering	
Course Code : 3359	Course Title: Fluid Mechanics Lab
Semester : 3	Credits: 1.5
Course Category: Programme Core Course (A)	
Periods per week: 3(L:3 T:0 P:3)	Periods per semester: 45

Course Objectives:

- perform experiments on flows through pipes and orifices.
- perform experiments on flows through channels and notches.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of Fluid Mechanics		Mechanics of Fluids	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Calibrate pressure gauges and verify Bernoulli's theorem.	8	Applying
CO2	Determine Hydraulic coefficient of orifices and friction coefficient of pipes.	9	Applying
CO3	Calibrate Flow-measuring devices.	12	Applying
CO4	Measure flow in Open-Channels.	12	Applying
	Lab Test	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Calibrate pressure gauges and verify Bernoulli's theorem.		
M1.01	Study of taps, valves, pipe fittings, gauges, Pitot tubes, water meters and current meters.	2	Understanding
M1.02	Calibration of Pressure gauges	3	Applying
M1.03	Use Bernoulli's apparatus to apply Bernoulli's theorem to get total energy line for a flow in a closed conduit of varying cross sections	3	Applying
CO2	Determine Hydraulic coefficient of orifices and friction coefficient of pipes.		
M2.01	Determination of Hydraulic coefficients of orifices under constant head method.	3	Applying
M2.02	Determine Time of Emptying through orifice	3	Applying
M2.03	Determination of friction co-efficient in pipes	3	Applying
	Lab Test -I	2	
CO3	Calibrate flow-measuring devices.		
M3.01	Calibration of Venturimeter	3	Applying

M3.02	Calibration of Orifice meter	3	Applying
M3.03	Calibration of water meter.	3	Applying
M3.04	Calibration of rectangular and triangular notches.	3	Applying
CO4	Measure flow in Open-Channels.		
M4.01	Determine Coefficient of discharge of Venturi flume	3	Applying
M4.02	Determine Manning's coefficient and Chezy's constant of channel	2	Applying
M4.03	Plot Specific Energy curve for channel flow	1	Applying
M4.04	Open Ended Experiment: Parameters of Hydraulic Jump in Open channels.	6	Applying
	Lab Test - II	2	

Text / Reference:

T/R	Book Title/Author
T1	Modi P. N. and S. M. Seth, Hydraulics & Fluid Mechanics, S.B.H Publishers, New Delhi, 2002.
T2	Subramanya K., Theory and Applications of Fluid Mechanics, Tata McGraw-Hill, 1993.
R1	Streeter.V.L. Fluid Mechanics, Mc Graw Hill Publishers.
R2	Frank.M.White, Fluid Mechanics, Mc Graw Hill, 2013.

Online resources:

Sl.No.	Website Link
1	https://www.nptel.ac.in/